MultiCore TECHNOLOGIES

LEARN TODAY BUILD TOMORROW

Your Future in Tech

SOFTWARE TESTING

MASTER PROGRAM

From Manual to Automation
Build Your Testing Expertise



COURSE HIGHLIGHTS

- Manual Testing & SDLC/STLC
- Selenium WebDriver with Java
- Core Java & OOP Concepts
- SQL & Database Testing
- TestNG, POM, JIRA & Automation Frameworks

SOFTWARE TESTING MASTER PROGRAM

From Manual Testing to Automation Excellence

ABOUT THE PROGRAM

Build a strong foundation in Software Testing, Core Java, Selenium Automation and SQL through a structured learning journey from manual testing fundamentals to advanced automation frameworks.

This program covers SDLC, STLC, Manual Testing, Test Case Design, Bug Reporting, Selenium WebDriver, Java Programming, TestNG, Page Object Model, SQL, Data-Driven Testing, Selenium Grid and Jenkins Integration.

KEY TOPICS & SUBTOPICS

1. **Manual Testing**
   - Testing fundamentals
   - Test processes
   - Defect management
2. **Core Java**
   - Java basics & OOP concepts
   - Programming essentials for automation
3. **Selenium with Java**
   - WebDriver fundamentals
   - Locators & actions
   - Test automation
4. **SQL & Database Testing**
   - SQL queries & joins
   - Data validation
   - Database testing concepts
5. **TestNG & Automation Frameworks**
   - TestNG features
   - Framework design
   - Parallel execution
6. **Page Object Model**
   - POM architecture
   - Reusable components
   - Maintainable framework
7. **Testing Tools & CI/CD**
   - JIRA, TestRail, Bugzilla
   - Jenkins Integration
   - Selenium Grid

WHO CAN JOIN?

- Students
- Graduates
- Beginners
- Working Professionals
- Developers
- Aspiring QA Professionals
- Software Testers

LEARNING OUTCOME

By the end of the program, learners will be able to design test cases, execute manual testing, report defects, automate web applications using Selenium with Java, perform database testing using SQL and build structured automation frameworks.

BUILD SKILLS
FOR A SAFER TOMORROW

MANUAL TESTING & CORE JAVA

TESTING | QUALITY | JAVA

1 MANUAL TESTING

INTRODUCTION TO SOFTWARE TESTING

- Introduction to Software Testing
- Importance of Testing in SDLC
- Functional & Non-Functional Testing
- Black-Box & White-Box Testing

LEVELS OF TESTING

- Unit Testing
- Integration Testing
- System Testing
- Acceptance Testing

TESTING METHODOLOGIES

- Waterfall Model
- Agile Methodology
- V-Model
- Spiral Model
- SDLC & STLC
- Role of Tester in Agile & Waterfall

TYPES OF SOFTWARE TESTING

Functional Testing

- Smoke Testing
- Sanity Testing
- Regression Testing
- Integration Testing
- UAT
- System Testing

Non-Functional Testing

- Performance Testing
- Load Testing
- Stress Testing
- Usability Testing
- Security Testing

Other Testing Types

- Compatibility Testing
- Ad-hoc Testing
- Exploratory Testing
- UI Testing

2 TEST CASE DESIGN & BUG MANAGEMENT

TEST CASE DESIGN

- Test Case Structure
- Test Case Writing
- Positive & Negative Test Cases
- Test Case Review
- Traceability Matrix
- Test Data Creation

BUG REPORTING & TRACKING

- Bug Life Cycle
- Defect Reporting
- Severity & Priority
- Steps to Reproduce
- Bug Verification & Closure

BUG TRACKING TOOLS

JIRA | Bugzilla | Redmine | Mantis

3 CORE JAVA

JAVA FUNDAMENTALS

- Java Syntax
- Variables & Data Types
- Operators
- Conditional Statements
- Loops
- Arrays

OBJECT-ORIENTED PROGRAMMING

- Classes & Objects
- Constructors & Methods
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Interfaces

EXCEPTION HANDLING

- Checked & Unchecked Exceptions
- try, catch & finally
- Custom Exceptions
- throw & throws

AI

BUILD INTELLIGENT SOLUTIONS

ANALYZE REAL-WORLD DATA

WORK WITH MODERN TOOLS

CREATE DATA-DRIVEN IMPACT

SKILLS TODAY
FOR A SMARTER TOMORROW

AVA, SELENIUM & TESTNG

JAVA & SELENIUM AUTOMATION

CORE JAVA ADVANCED CONCEPTS

4 Collections Framework

- List
- Set
- Map
- Queue
- Iterators
- Enhanced For Loop
- Sorting & Searching Collections

Multithreading

- Thread Class
- Runnable Interface
- Synchronization
- Thread Communication
- Deadlock Handling
- Executor Framework
- Thread Pools

Java I/O

- File Handling
- FileReader & FileWriter
- BufferedReader & BufferedWriter
- Input & Output Streams
- Serialization & Deserialization

Java 8 Features

- Lambda Expressions
- Functional Interfaces
- Stream API
- Filtering & Mapping
- Reducing Collections
- Optional Class

JUnit

- Unit Testing with JUnit
- Test Annotations
- Assertions
- Test Execution

5 SELENIUM WITH JAVA

Introduction to Selenium

- Selenium Fundamentals
- Selenium IDE
- Selenium WebDriver
- Selenium Grid
- Selenium WebDriver vs Selenium RC
- Advantages of Selenium

Selenium WebDriver Setup

- Java Installation
- Eclipse / IntelliJ Setup
- Selenium WebDriver Setup
- WebDriver Configuration
- Creating Basic Selenium Scripts

Locating Web Elements

- ID
- Name
- ClassName
- TagName
- LinkText
- PartialLinkText
- XPath
- CSS Selectors

WebDriver Commands

- get()
- navigate()
- maximize()
- close()
- quit()
- click()
- sendKeys()
- clear()
- submit()

Synchronization & Waits

- Implicit Wait
- Explicit Wait
- Fluent Wait

Browser & Web Elements

- Alerts & Popups
- Dropdowns
- Frames & iFrames
- Multiple Windows & Tabs
- Cookies
- Screenshots
- File Upload & Download

6 TESTNG FRAMEWORK

TestNG Fundamentals

- TestNG Introduction
- Annotations
- Assertions
- Test Execution

Test Organization

- @Test
- @BeforeMethod
- @AfterMethod
- Test Groups
- Test Priorities
- Parallel Execution

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ADVANCED SELENIUM & AUTOMATION FRAMEWORK

7 PAGE OBJECT MODEL — POM

POM FUNDAMENTALS

- Introduction to Page Object Model
- Advantages of POM
- Page Object Classes
- Web Element Management
- Reusable Methods

POM FRAMEWORK DESIGN

- Creating POM Classes
- Designing Test Classes
- Maintaining Test Scripts
- Reusable Components
- Scalable Automation Structure

8 ADVANCED SELENIUM CONCEPTS

DYNAMIC WEB ELEMENTS

- Dynamic XPath
- Dynamic Web Elements
- AJAX Applications
- JavaScript Pop-ups
- Handling Dynamic Content

BROWSER AUTOMATION

- Multiple Windows & Tabs
- Frames & iFrames
- Browser Cookies
- Screenshots
- File Upload & Download

SELENIUM GRID

- Selenium Grid Fundamentals
- Distributed Test Execution
- Cross-Browser Testing
- Parallel Execution



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ANALYZE REAL-WORLD DATA

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CREATE DATA-DRIVEN IMPACT

SKILLS TODAY
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9 DATA-DRIVEN TESTING

DATA-DRIVEN AUTOMATION

- Data-Driven Testing Concepts
- Excel-Based Testing
- Apache POI
- Reading Excel Data
- Writing Test Data
- Parameterization
- Multiple Test Data Sets

AUTOMATION TEST EXECUTION

- TestNG Integration
- Parallel Test Execution
- Test Suite Management
- Test Reports

10 TEST ENVIRONMENT & TESTING TOOLS

TEST ENVIRONMENT

- Test Environment Setup
- Environment Configuration
- Test Data Configuration
- Configuration Management

TEST MANAGEMENT TOOLS

- TestRail | HP Quality Center

DEFECT MANAGEMENT

- JIRA | Bugzilla | Redmine | Mantis

SQL, CI/CD, PROJECTS & CAREER

11 SQL & DATABASE TESTING

SQL FUNDAMENTALS
- Relational Database Concepts
- Tables, Rows & Columns
- Primary & Foreign Keys
- Relationships
- Database Schema

BASIC SQL QUERIES
- SELECT
- WHERE
- AND & OR
- LIKE
- IN
- BETWEEN
- IS NULL
- ORDER BY
- ASC & DESC
- LIMIT & OFFSET

DATA AGGREGATION
- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()
- GROUP BY
- HAVING

JOINS
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- SELF JOIN
- CROSS JOIN

DATA MANIPULATION
- INSERT
- UPDATE
- DELETE
- TRUNCATE
- COMMIT
- ROLLBACK
- SAVEPOINT

ADVANCED SQL
- Subqueries
- Nested Queries
- CASE WHEN
- Window Functions
- ROW_NUMBER()
- RANK()
- PARTITION BY

DATABASE OBJECTS
- Indexes
- Views
- SQL Functions
- String Functions
- Date & Time Functions
- Mathematical Functions

12 CONTINUOUS TESTING & CI/CD

JENKINS INTEGRATION
- Jenkins Fundamentals
- Selenium & Jenkins Integration
- Automated Test Execution
- Continuous Testing
- Build & Test Automation

AUTOMATION WORKFLOW
CODE → BUILD → AUTOMATED TESTING → TEST REPORTS → CONTINUOUS INTEGRATION

13 HANDS-ON PROJECTS

BUILD REAL-WORLD SOFTWARE TESTING PROJECTS

E-Commerce Application Testing
Functional, regression and UI test cases for an e-commerce application.

Selenium Automation Project
Automated web test cases using Selenium WebDriver with Java and TestNG.

Database Testing Project
Validate application data and perform database testing using SQL queries and joins.

POM Automation Framework
Develop a maintainable Selenium automation framework using Page Object Model.

End-to-End Testing Project
Complete testing from test case design and execution to defect reporting and automation.

14 SKILLS YOU WILL MASTER
- Manual Testing
- SDLC & STLC
- Core Java
- Selenium WebDriver
- TestNG
- Page Object Model
- SQL
- Database Testing
- Test Case Design
- Bug Reporting
- JIRA
- Data-Driven Testing
- Selenium Grid
- Jenkins
- Automation Frameworks

15 CAREER OPPORTUNITIES
- Manual Tester
- Software Test Engineer
- QA Engineer
- Automation Test Engineer
- Selenium Automation Tester
- Quality Assurance Analyst
- Test Automation Developer
- Software QA Analyst
- SDET
- QA Automation Engineer

BUILD REAL-WORLD PROJECTS

PRACTICAL TOOLS & TECHNOLOGIES

HANDS-ON LEARNING

CAREER GROWTH

SKILLS TODAY FOR A SMARTER TOMORROW